# MPA Short Course

# 4. MPA Identification and regulation

UK International Development
Partnership | Progress | Prosperity

Developed by the Ocean Country Partnership Program for the University of Cape Coast

# MPA Course Overview

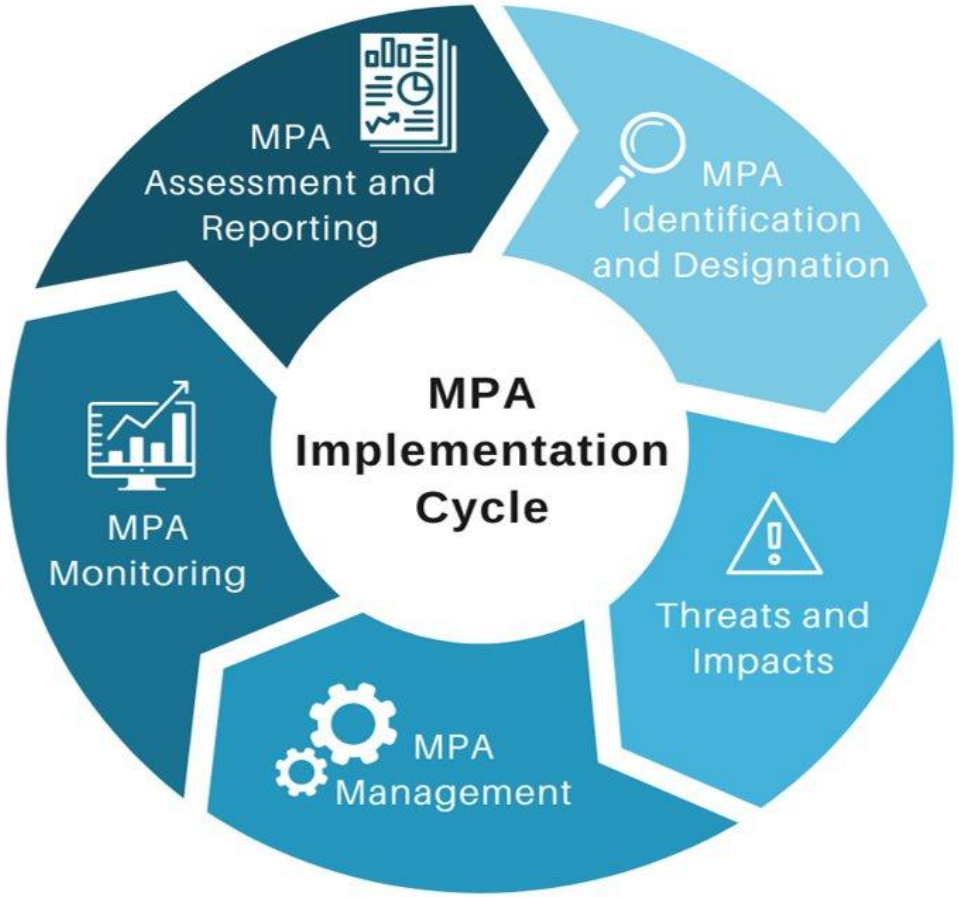
1. ~~Introduction to MPAs~~ ✓
2. ~~MPAs around the world~~ ✓
3. ~~MPAs in Ghana~~ ✓
4. **MPA identification and regulation** (You are here!)
5. MPA management planning & management
6. MPA monitoring
7. MPA & Marine Spatial Planning
8. Your Voice

For more information about this course, please contact sika.abrokwah@ucc.edu.gh

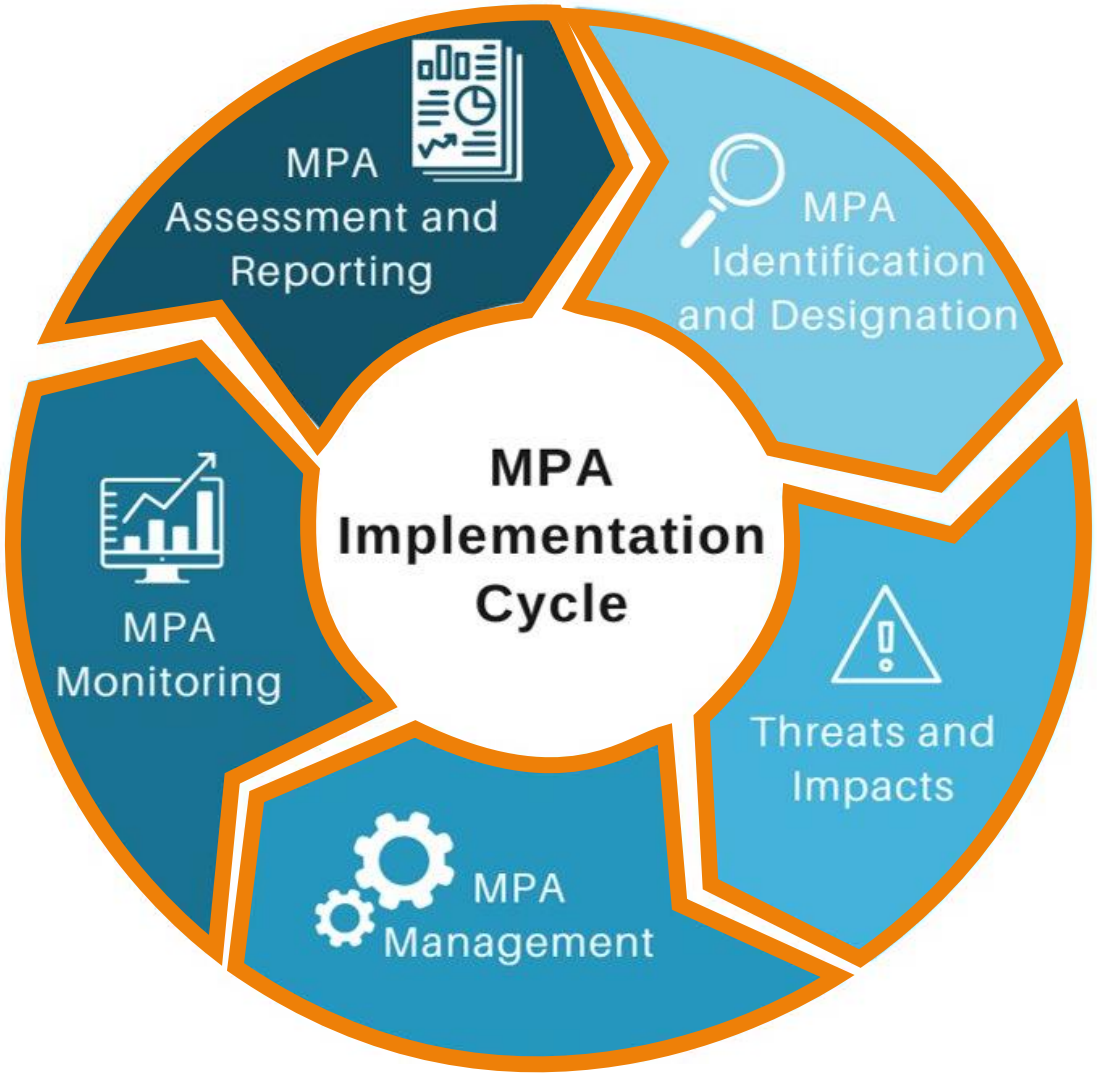
# Learning Objectives

**In this module you will:**

- Become familiar with the MPA Implementation cycle
- Recognise the importance of stakeholder engagement
- Consider steps for MPA identification and selection
- Further understand the benefits and principles of MPA networks



# MPA Implementation Cycle



# Legislation Document

- MPA Designation
- MPA Objectives and Purpose
- MPA Zoning and Regulations
- Enforcement and Compliance

- Stakeholder Engagement / Participation
- Funding and Resources
- Monitoring
- Evaluation

# Importance of Stakeholder Engagement

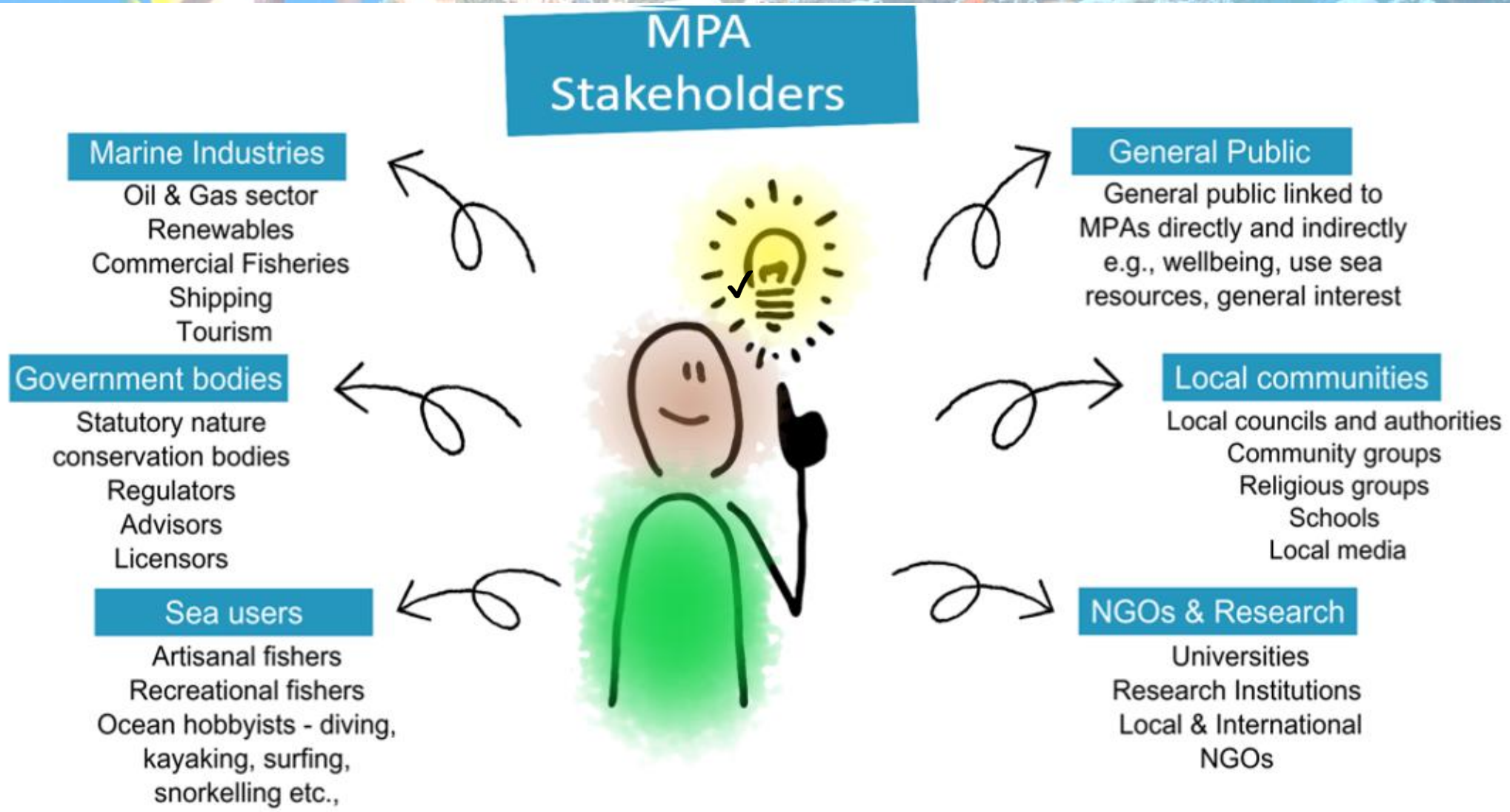
**Stakeholder engagement is the most important factor affecting MPA success, and its absence is the most important factor influencing failure[1]**

- Raising awareness and understanding the need for MPA governance
- Collecting and sharing information
- Gathering views on proposed management
- Sharing decision-making (including advisory groups)
- Ensuring stakeholder buy-in
- Reduce conflict and mistrust

Stakeholder engagement with fishers and Fisheries department, Belize.

[1]Giakoumi, S., McGowan, J., Mills, M., Beger, M., Bustamante, R.H., Charles, A., Christie, P., Fox, M., Garcia-Borboroglu, P., Gelcich, S. and Guidetti, P. (2018). Revisiting "success" and "failure" of marine protected areas: a conservation scientist perspective. Frontiers in Marine Science, 5: 223

# Who are the stakeholders?



# Principles of MPA Design

## Define objectives & goals of the site

- Threatened species/ critical habitats/ ecosystems
- Breeding, recruitment and nursery grounds
- Protect cultural values e.g., areas of recreation
- Maintain or restore

# SMART Objectives

Specific

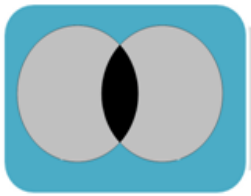
**S** Specific objectives provide **clarity and precision** in defining the specific conservation targets of an MPA.

Measureable

**M** Measurable objectives allow for the use of **quantifiable indicators and metrics** to assess progress and success.

Attainable

**A** Attainable objectives are **realistic and achievable**.



Relevant

**R** Relevant objectives **align** with **broader conservation strategies**, **policies**, and **priorities**.

Time Based

**T** Time-based objectives set a **clear timeline** for achieving the desired outcomes.

# SMART Objectives

1. To conserve mangrove forest habitats in the coastal zone through the creation of a Marine Protected Area..

2. To increase live coral cover by 25% within five years through the establishment of a Marine Protected Area in the designated coastal reef habitat.

3. To restore degraded seagrass beds to their original ecological condition through the implementation of a Marine Protected Area in the study area.

# SMART Objectives

1. ✓ Specific
   ✓ Relevant
   ✕ Not measurable
   ✕ Not time-bound
   ✓ Attainable

2. ✓ Specific
   ✓ Measurable
   ✓ Attainable
   ✓ Relevant
   ✓ Time-bound

3. ✓ Specific
   ✓ Measurable
   (implied restoration outcome)
   ✓ Relevant
   ✕ Not clearly attainable
   ✕ Not time-bound

# MPA Conservation Objectives: Trajectories

## Restore / Recover

DIRECT EVIDENCE OF DAMAGE

EXPOSURE TO PRESSURES AS A PROXY OF CONDITION

RESTORATION ACTIVITIES

'DEGRADED'

'HEALTHY'

MANAGEMENT INTERVENTION – REDUCE/REMOVE PRESSURES

# Principles of MPA Design

## Scientific basis

- Best available evidence
- Biodiversity surveys, habitat mapping etc

## Ecological representation

- Protect areas that best represent the range of habitats and species in the region

## Connectivity

- Links to other MPAs and ecologically important areas

# Principles of MPA Design

## Climate change

- Assess vulnerability to climate change impacts
- Identify key climate change threats
- Select resilient sites

## Boundary

- Guided by conservation objectives
- Distribution of habitats / movement patterns of species
- Are there natural geographic features to help delineate the site?
- Buffer zones

# Principles of MPA Design

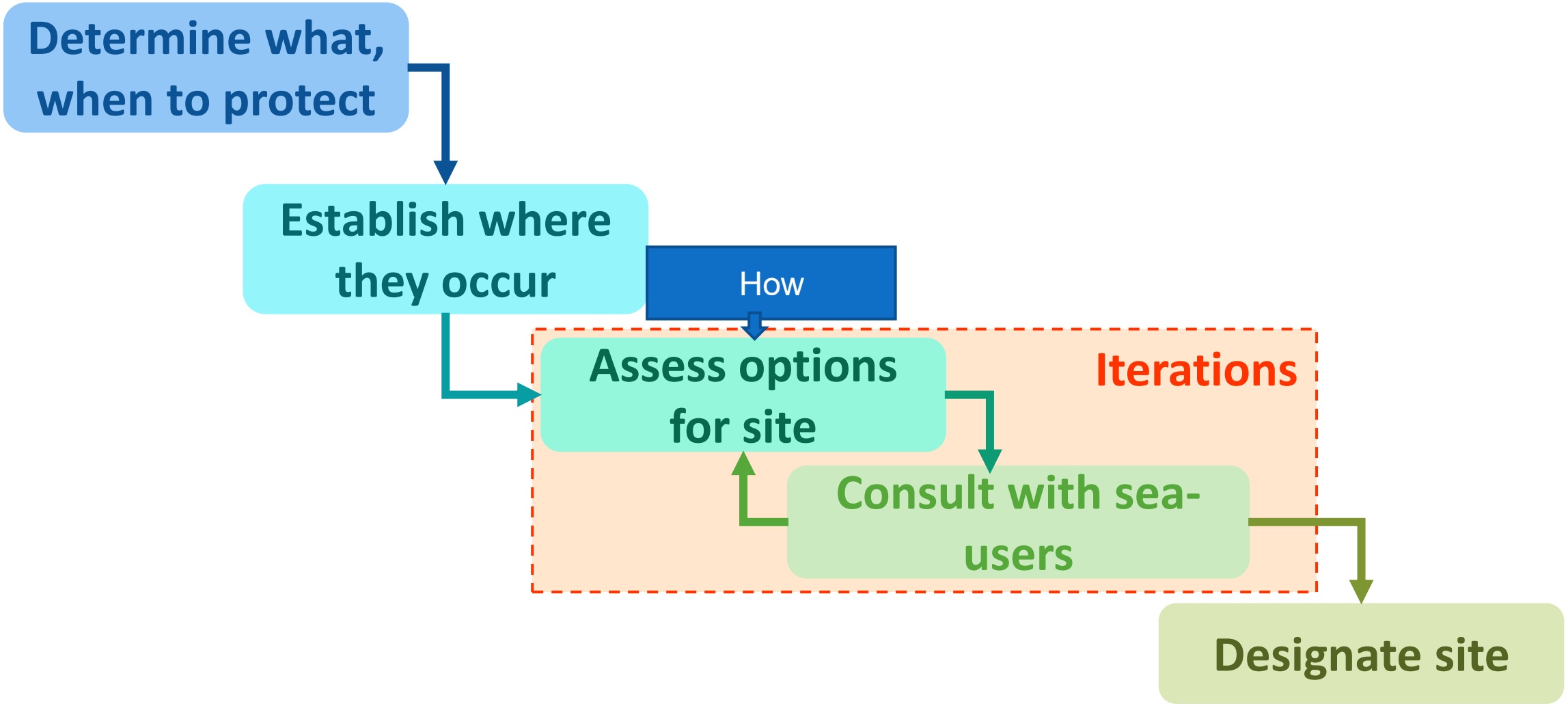
## Stakeholder engagement

- How is the site currently used – feasibility of success
- What are the current threats?
- Socioeconomic impacts

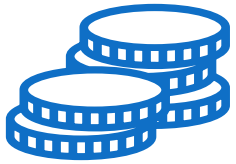
## Legislation

- What regulation can the site be designated under
- Take into account existing management

# Designation Process Summary



# Challenges to Designation



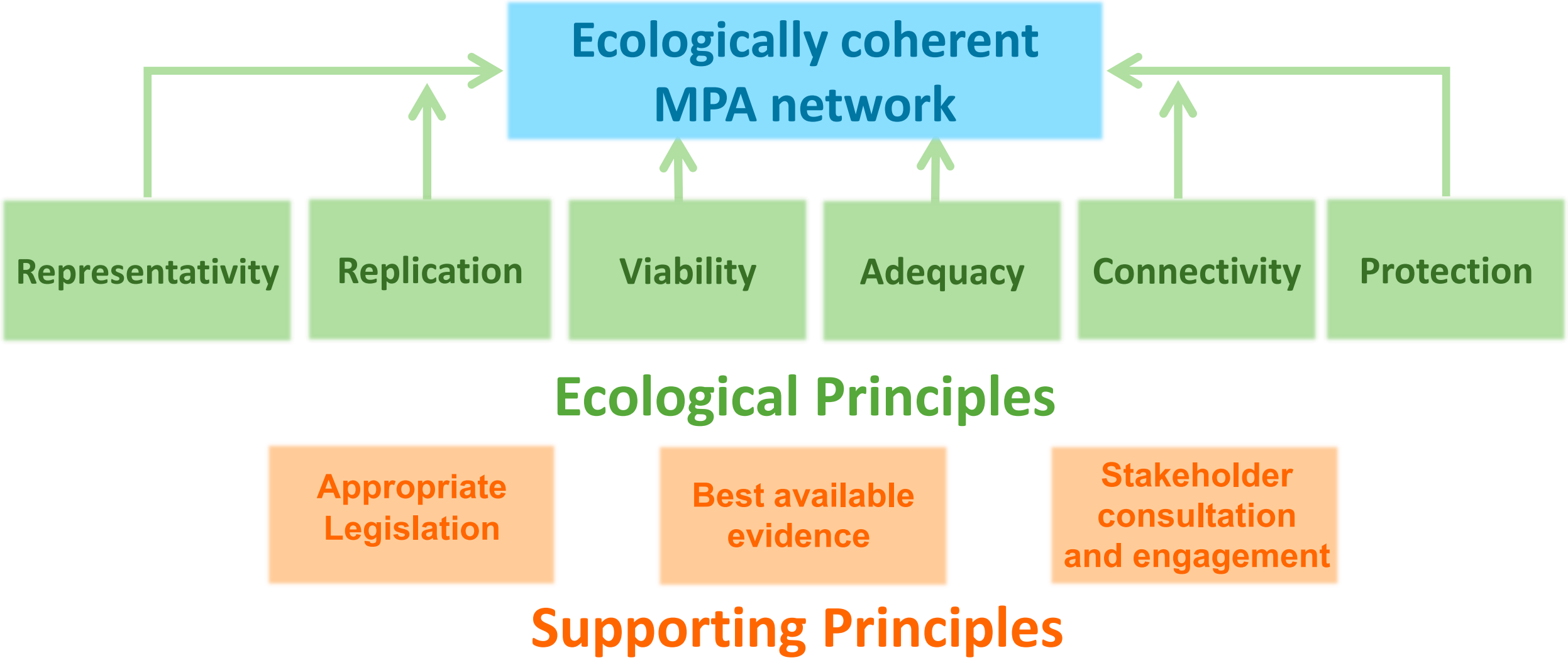
**Resources –** staff requirements, data collection and stakeholder engagement costs

**Uncertainty -** What is the most effective management measure? How do we measure progress?

**Lack of evidence -** Knowledge and data gaps. Recognising data and evidence limitations.

**Stakeholder attitudes -** Public support or objection to plans can influence success.

# MPA Networks: Key Principles



# Ecological Principles

**Representation -** represent the range of marine habitats and species.

**Replication -** all major habitats should be replicated and distributed throughout the network.

**Viability -** incorporate self-sustaining, geographically dispersed component sites of sufficient size.

# Ecological Principles

**Adequacy –** Adequate size to deliver its ecological objectives and ensure the long-term protection and/or recovery of populations, species and communities .

**Connectivity -** maximises and enhances the linkages among individual MPAs using the best science available.

**Protection -** the MPA network is likely to include a range of protection levels, ranging from highly protected sites to areas with minimal restrictions on activities.

# Supporting Principles

**Appropriate legislation -** A comprehensive legal framework is essential to provide a solid foundation for MPAs, ensuring their functionality and effectiveness.

**Best available evidence -** Network design should be based on the best information currently available.

**Stakeholder consultation & engagement –** Involving different stakeholders in all stages of MPA network implementation ensures their knowledge and support are included, and therefore buy-in.

# Summary

**You should now:**

- Understand the MPA implementation cycle
- Be aware of why stakeholder engagement is important
- Understand the theory of MPA identification and design
- Understand the need to develop networks of MPAs
- Be familiar with the key principles of MPA networks